ISMB 2018

PREDICTING NEW RELATIONSHIPS BETWEEN GENES AND HPO TERMS

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BIH-JRG Computational Genome Biology

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Motivation

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Patient with genotypes and phenotypes

- No mutation in known disease associated genes
  - 17% genes are annotated in human disease*

How to assign possible phenotypes to unknown genes?

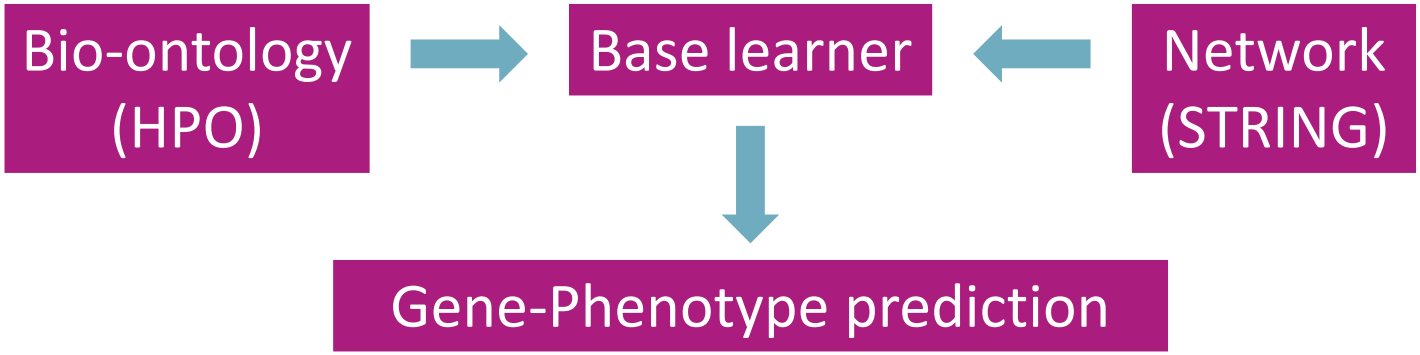
➡ Using gene networks, e.g.

  - Protein-Protein interaction
  - Gene expression
  - Text-mining

- Learn new relations by machine-learning

* HPO 06/2018: 3740 genes linked to disease
Ensemble v92: 21475 genes

Introduction



Flat Classifier

- Predicts each class/term separately
- Hierarchy-unaware
  - Neglects structural information between classes
  - Violation of hierarchical constraints

→ inaccurate predictions

Available software packages

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- PHENOstruct [1]
  - Good performance
  - Slow
  - Fixed base learner (structural support vector machine)
- Clus-HMC-Ens [2]
  - Hierarchical Multilabel Classification (HMC)
  - Fixed base learner (decision tree)

[1] I. Kahanda, et al. PHENOstruct: Prediction of human phenotype ontology terms using heterogeneous data sources. F1000Research. 2015;4
[2] C. Vens, et al. Decision trees for hierarchical multi-label classification. Machine Learning. 2008;*73*(2),185

HEMDAG software package

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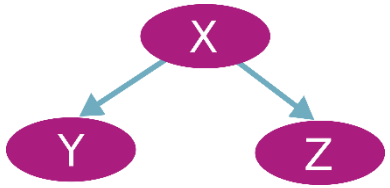
- Flexible input:

Ontology (HPO)



Annotations

Genes	HPO-Terms: Z	X	Y
A	1	1	1
B	0	1	1
C	0	0	0

Network (STRING)

Genes	Genes: A	B	C
A	0.0	0.2	0.9
B	0.2	0.0	0.3
C	0.9	0.3	0.0

Flat prediction

Genes	HPO-Terms: Z	X	Y
A	0.9	1.0	0.9
B	0.1	1.0	0.8
C	0.6	0.7	0.7

- Works with any flat predictor
- Multiple algorithms to make the predictions hierarchical aware
- Integrated optimization
- Tutorial: http://hemdag-tutorials.rtfd.io

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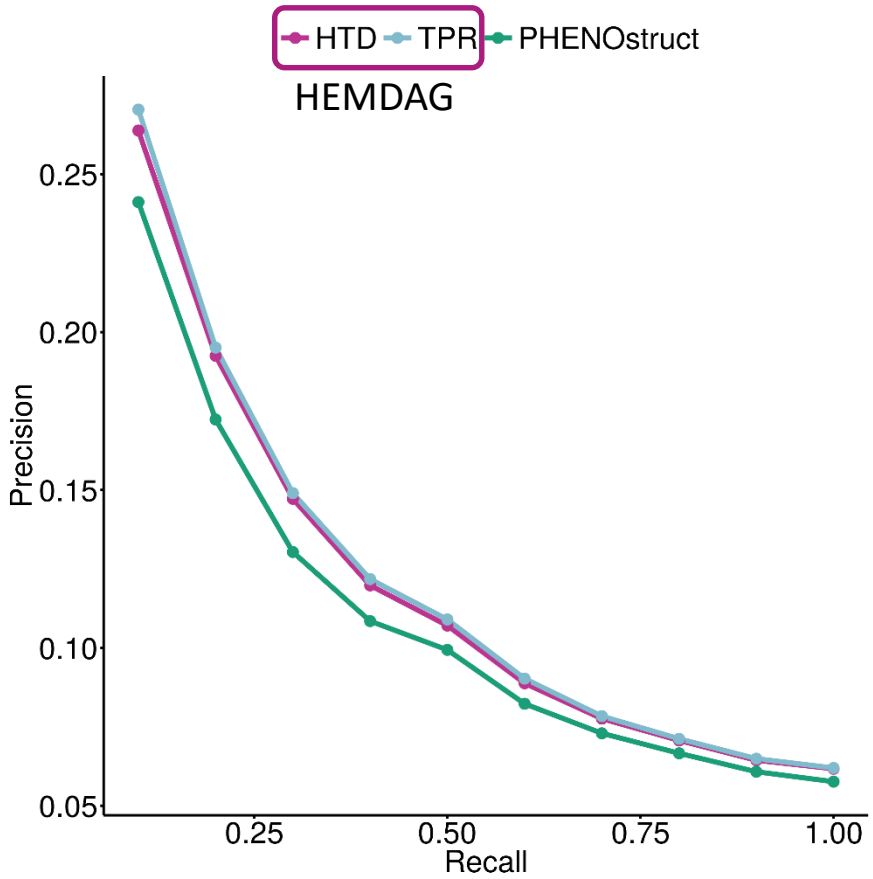
Results



- Base learner: SVM
- Training
  - HPO 04/2016
  - STRING 9.1
- Testing:
  - HPO 03/2017
- Fast (up to 20 min)

POSTER B-579; Contributors and Funding

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Computational Biology and Bioinformatics

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- Additional, detailed algorithm description
- Extended performance evaluation
- Software package HEMDAG
- Tutorial

http://hemdag-tutorials.rtfd.io

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